

Theodore Mollano

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RESEARCH SUMMARY

My research is in applied particle physics, where intuition, numerics, and careful experimentation are all necessary to understand system dynamics. My approach is distilled by the advice of my professor, Ivo Terek: *“in a calculation, you always want to know where you need to go before you put your pen to paper.”*

EDUCATION

Williams College Sep 2021 – Present

B.A, Physics & Mathematics with Honors, Williamstown, MA

Major GPA: 3.96; Dean's List every term. Took medical leave in Fall 2022 from ski accident

Dual honors thesis in both Mathematics and Physics under Dr. Terek, and Dr. Giovanetti.

University of California Berkeley Feb 2024 – May 2024

Mathematics Study Program, Berkeley, CA.

Grad coursework in Mathematics for a designated study-away program.

Budapest Semesters in Mathematics May 2023 – Aug 2023

Mathematics Summer Program, Budapest, Hungary.

Mathematics coursework and research for a designated study-away program.

RESEARCH EXPERIENCE

Physics Research Assistant, Giovanetti Lab, Williams College 2022 – 2025

- Built dewar and LN-outgassing system for in-lab testing for DARKSIDE 20K experiment in Gran Sasso, Italy.
- For thesis, calculated half life of $^{180\text{m}}\text{Ta}$ with data from MAJORANA Demonstrator experiment in South Dakota.

Mathematics Research Assistant, Williams College 2025

- For thesis, classified moment μ_c images for the Lie-group action of $\text{SO}(3)$, supervised by Prof. Ivo Tereurk.
- Research in plane tiling dynamics in Ergodic Theory with Professor Cesar E. Silva. Ongoing.

Chemical Graph Intern, Mathematica (Remote/In-Person) 2022 – 2023

- Created algorithms for molecular structure recognition for isomer enumeration. Results published.
- Developed software for chemical synthesis prediction in Mathematica in Chemistry team.

TEACHING EXPERIENCE

Physics/Mathematics Teaching Assistant, Williams College 2023 – 2025

- Taught Electromagnetism, Abstract Algebra, Measure Theory, Linear Algebra and Dynamics for 10 hrs/week.

Mathematics Teaching Assistant, AwesomeMath 2022

- Delivered intermediate combinatorics instruction to a cohort of summer students.

HONORS

2021–Present: Physics Major with Honors, Mathematics Major with Honors, Dean's List every term.

TECHNICAL SKILLS

Programming: Python, C++, Machining, Metalworking, Finite Element Simulation of 2D/3D PDEs.

Languages: English (Native), German (Professional), Italian (Intermediate), Chinese (Beginner).

TALKS

Setting Limits on the Tantalum Lifetime, Williams College

Jul 2025

- Presented findings on the half-life calculation of $^{180\text{m}}\text{Ta}$ using a likelihood-fit on data from the MAJORANA Demonstrator experiment.
- Discussed Monte-Carlo simulations (in Geant4) used to calculate geometric detector efficiency.

A Delzant-Type Correspondence For Hamiltonian $\text{SO}(3)$ actions, Williams College

Apr 2025

- Mathematics Thesis Defense. Characterized symmetry on a manifold M by a Lie group action $G \curvearrowright M$, relating to Nöther's Theorem and conserved quantities.
- Classified moment μ_c images for the Lie-group action of $\text{SO}(3)$ as either a ball or slicings of a ball and presented the findings, as an analogue of Delzant's Theorem.

An Introduction to Isomer Generation with Wolfram Language, Champaign-Urbana

Dec 2022

- Presented a talk on k -valent graphs and wrote a corresponding outreach article on graph-based algorithms for chemical isomer generation, benchmarking implementations against the SURGE structure generator.
- Completed this project under the mentorship of Dr. Robert Nachbar (Wolfram), focusing on communicating computational chemistry methods to a general technical audience.